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Implementation science frameworks for active and sustainable transportation interventions with a focus on equity: A scoping review

Emily McCullogh [a,*], Sarah A. Moore [b], Rebecca Maher [c], Nazanin Jannati [d], Maud Muosieryiri [d], Alison Macpherson [a], Ben Beck [e], Daniel Fuller [d]

[a] York University, 4700 Keele Street, Toronto, Ontario M3J1P3, Canada
[b] Dalhousie University, 6885 University Avenue, Halifax, Nova Scotia B3H4H6, Canada
[c] Memorial University of Newfoundland, 300 Prince Philip Dr, St. John's, Newfoundland A1B3V6, Canada
[d] University of Saskatchewan, 110 Science Place, Saskatoon, Saskatchewan S7N5C9, Canada
[e] Monash University, 553 St. Kilda Road, Melbourne, Victoria 3004, Australia

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ABSTRACT

Background: There is a growing need to promote sustainable transportation in urban environments. Active and sustainable transportation interventions are implemented through a pathway of actors responsible for funding, design, planning, and construction. This scoping review summarizes peer-reviewed literature that uses implementation science theories, models, and/or frameworks for active and sustainable transportation interventions and categorizes the included frameworks using an existing implementation science taxonomy. Barriers and facilitators to implementation are identified, as well as the inclusion of transportation equity considerations in implementation, if present.

Methods: Six databases were searched from inception to June 4, 2025. Records were screened at title, abstract, and full-text phases by two reviewers independently. Conflicts were resolved by consensus or with a third reviewer. Articles were summarized using thematic analysis and narrative synthesis.

Results: Searches resulted in 8892 records; 8561 after duplicates were removed. At the title and abstract phase 6561 articles were screened and 114 were screened at full-text phases. 26 articles met inclusion criteria. Factors operating as both facilitators and barriers include context-specificity, community engagement and participation, partnerships and collaboration, and funding. Equity was addressed in eight articles.

Conclusions: Implementation science frameworks utilized for active and sustainable transportation interventions in urban environments exist; however, attention to transportation equity is limited. Frameworks should be tailored to the context and flexible enough to adapt to local settings. It is recommended that equity be embedded throughout the implementation process, including meaningful engagement with communities, to promote active transportation and address local mobility needs.

1. Introduction

In a rapidly growing and urbanized world we are facing health challenges due to physical inactivity, chronic diseases, and road traffic injury (Beaglehole et al., 2012; Wen et al., 2013), exacerbated by climate change (Giles-Corti et al., 2016). Globally, the urban population is expected to grow by approximately 2.5 billion by 2050 (Ramaswami et al., 2016), signifying substantial increases in the number of city dwellers and need for solutions that tackle these challenges (Brown et al., 2010). Roughly 60% of the urban areas that will exist in 2050 have yet to be constructed, indicating substantial demands for new infrastructure (United Nations, 2013). Meanwhile, cities across the globe are facing demands of aging infrastructures, requiring urgent maintenance and replacement (United Nations, 2013). Thus, there is an urgent need reimagine the design of our cities, which includes the infrastructure to facilitate travel and transport.

A promising way forward is to apply tools and strategies from other scholarly areas to expand our knowledge and imagine new possibilities,

* Corresponding author.
E-mail addresses: emilym03@yorku.ca (E. McCullogh), sarah.moore@dal.ca (S.A. Moore), ram424@mun.ca (R. Maher), nazanin.j@usask.ca (N. Jannati), mmuosieyiri@umass.edu (M. Muosieryiri), alison3@yorku.ca (A. Macpherson), ben.beck@monash.edu (B. Beck), daniel.fuller@usask.ca (D. Fuller).

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which is the motivation for this scoping review that highlights key linkages between active and sustainable transportation interventions and implementation science. Further, we emphasize the importance of accounting for equity in transportation planning and implementation. This scoping review is guided by the following research questions: Do implementation science frameworks for active and sustainable transportation interventions exist? If so, what key barriers and facilitators are identified? and how is equity considered and operationalized?

To answer these questions, this scoping review synthesizes current literature on implementation science frameworks and their application in active and sustainable transportation interventions, broadly, using narrative synthesis (Popay et al., 2006; Rodgers et al., 2009); specifically, the synthesis will highlight key barriers and facilitators to implementation, and examine *if* and *how* transportation equity is acknowledged and/or incorporated in the frameworks provided. Unlike traditional implementation science studies examining a single intervention, our review purposefully includes a broad range of interventions relevant to active and sustainable transportation to provide an overview of the current landscape. Lastly, to situate included frameworks within an existing taxonomy we utilized Nilsen's (2015) categories of implementation science theories, models, and frameworks to contextualize our results (Table 1).

This scoping review is an integral component of CapaCITY/É, a research programme designed to adapt and apply implementation science in the realm of healthy and equitable cities, focusing on active and sustainable transportation interventions (Winters et al., 2024).

2. Background information

2.1. Active and sustainable transportation

In response to global health (physical inactivity, chronic disease, road traffic injury) and environmental (climate change) challenges, there has been increased scholarly (Bland et al., 2024; Jelti et al., 2023; Toledano et al., 2025) and organizational (Transport Canada, 2019; United Nations, 2023; World Health Organization, 2026) prioritization of active and sustainable transportation, which includes programs and strategies to encourage active travel (Batool et al., 2024; Pan & Ryan, 2025), as well as significant changes to the built environment and transportation infrastructure (Kapur et al., 2021; Kopal & Wittowsky, 2023; Laberee et al., 2023). We acknowledge that "active and sustainable transportation" is a broad concept encapsulating many forms of travel and infrastructure changes. As defined by the United Nations (2025), sustainable transportation "refers to mobility systems that seek to minimize greenhouse gas emissions and environmental impacts, while ensuring safety and affordability, improving energy and resource efficiency, and providing equitable access to mobility for all" (para. 1). This definition includes public transport and electric vehicles, as well as walking and cycling. However, while the "active" dimension of sustainable transportation is implied, we assert its importance in the context of this research and will continue to use "active and sustainable transportation" to identify the scope of this review.

Active and sustainable transportation plays a vital role in creating and maintaining healthy cities (Banister, 2008; Nieuwenhuijsen, 2020), particularly through the provision of infrastructure and interventions that support safe physical activity for city dwellers (Giandonato et al., 2021; Keall et al., 2022; Morgan, 2013; Timmons et al., 2024). In other words, transport can influence people's quality of life via "impacts on travel distance, time, cost, comfortability of daily mobility, and living amenity" (Wan & Titheridge, 2024, p. 4). Examples of interventions include building cycling infrastructure (Rissel et al., 2013), pedestrianizing areas and reducing traffic speeds (Castillo-Manzano et al., 2014), implementing car-sharing programs (Ciari et al., 2008), adopting transit-oriented development principles (Berawi et al., 2019), and expanding public transit (Fesselmeyer & Liu, 2018). Further, links between sustainable transportation infrastructure and people's physical and mental health are well established (Elsamani & Kajikawa, 2024; Mundorf et al., 2018; Tinella et al., 2023); specifically, infrastructure that supports cycling (Garrard et al., 2012; 2021; Pucher & Buehler, 2012) and walking (Kelly et al., 2018; Roe et al., 2020), which facilitates active travel (Aldred, 2019; Timmons et al., 2024) and is a key component of healthy cities (Winters et al., 2017). There is also a growing body of research on sustainable transportation, broadly, which is critical to healthy cities (de Leeuw et al., 2022; Siri, 2016; World Health Organization, 2023; Zhao et al., 2020). Despite this progress, implementing changes to urban infrastructure to support the health and safety of diverse and growing city populations remains challenging.

2.2. Implementation science.

Tools and strategies employed in *implementation science* can be utilized to address these concerns. As a relatively new discipline, implementation science aims to systematically investigate methods, strategies, factors, and processes that influence the integration of evidence-based interventions into practical settings (Lobb & Colditz, 2013). Originating in the realm of healthcare to assess health outcomes in clinical, organizational, public health, and/or policy settings (Wilson et al., 2024), the scope of implementation science application is traditionally contained to a specific health policy or program (Eccles & Mittman, 2006). Of particular interest is the study of barriers and facilitators to implementing heath interventions (e.g., Richmond et al., 2020; van der Laag et al., 2023). However, little research has sought to develop and use implementation science frameworks for active and sustainable transportation interventions (Smith, 2018) because existing frameworks tend to operate at the individual or organizational levels, often oversimplifying factors that operate at the community, policy, or societal levels (McIsaac et al., 2018). Further, existing frameworks do not accommodate the complexity of the types of intersectoral solutions required for sustainable transportation interventions (Pettersson & Hrelja, 2020), nor do they have the ability to determine effectiveness of a single intervention (e.g., bike lanes) relative to a bundle of interventions (e.g., mobility strategies) (Bao et al., 2015; Parachute, 2022).

While implementation science and its tools were not originally designed to examine complex and expansive active and sustainable transportation interventions, we contend that it is an appropriate lens that aids in the advancement of understanding *why* and *how* such interventions are successfully implemented, or not. For example, the study of barriers and facilitators is particularly useful in the context of built environment changes for active and sustainable transportation (McCullogh et al., 2022; Northridge et al., 2019), which can assist decision- and policy-makers in making such changes in their cities. Further, as a multidisciplinary approach, implementation science takes into account the social, political, economic, and cultural contexts (Baumann & Cabassa, 2020), enabling a better understanding of *how* interventions can be effectively adapted to specific settings (Lobb & Colditz, 2013; Moore, 2023; Nilsen & Bernhardsson, 2019). Thus, utilizing implementation science tools in the context of active and sustainable transportation interventions invites an opportunity to expand the scope of implementation science and enhance its utility beyond the scope of healthcare.

2.3. Equity

An additional challenge with employing implementation science tools to examine active and sustainable transportation interventions is that they often lack attention to equity, which is a critical consideration in the implementation of interventions to support healthy cities (Brownson et al., 2021; Martens et al., 2019; McCullogh et al., 2023). In the context of transportation, equity "encompasses the fair distribution of transportation resources, inclusive participation in decision-making processes, and recognition of the prevailing injustices that shape different levels of need and power within transportation systems" (Williams et al., 2023, p. 7). While researchers have examined current policies and practices with the goal of advancing sustainable transportation equity (Martens et al., 2019; Sanguinetti & Alston-Stepnitz, 2023; Williams et al., 2023), there are still challenges to measuring equity in this context (Guo et al., 2020; LeClair et al., 2023). The results of this scoping review will contribute to efforts towards addressing these challenges by identifying existing research utilizing implementation science frameworks in the context of active and sustainable transportation interventions, and illustrating *how* equity has been included, or not.

3. Methods

This scoping review was performed in compliance with Arksey and O'Malley and the Joanna Briggs Institute (JBI) methodology (Arksey & O'Malley, 2005; Khalil et al., 2020; Maher et al., 2020; Peters et al., 2020), with the checklists for the Preferred Reporting Items for Systematic review and Meta-Analysis extension for Scoping Reviews (PRISMA-ScR), and the Preferred Reporting Items for Systematic review and Meta-Analysis Protocols (PRISMA-P) (Tricco et al., 2018). A protocol for this review was registered on Open Science Framework (osf.io/jwz3r) on June 20, 2022, and updated on December 5, 2023 (Maher et al., 2020).

3.1. Search strategy

We sought literature that described implementation science theories, models, and/or frameworks and their application to active and sustainable transportation. In implementation science, a theory explains the causal mechanisms of implementation, models are generally used to guide the translation of research into practice, and frameworks are more descriptive, focusing on factors influencing implementation (Damschroder, 2020; Nilsen, 2022). A librarian-assisted search was performed on July 26, 2022 and updated on June 4, 2025. First, a limited search of Scopus was performed to identify articles broadly on the topic. Second, words contained in the titles, abstracts, and index of these articles were subsequently used to develop the comprehensive search strategy and seven databases (Scopus, Medline, Web of Science, Sociological Abstracts, Case Index, Greenfile, and Environmental Science) were searched from database inception (see example search strategy in Appendix A). Third, after article selection, reference lists were also screened in duplicate to identify additional relevant articles.

3.2. Evidence selection

Evidence selection was guided by five inclusion criteria: 1) an available abstract; 2) the language of the paper (English or French only); 3) the paper was original research; 4) the paper was peer-reviewed; 5) the study used or described an implementation science theory, model or framework; and 6) the study applied the theory, model or framework to an active and sustainable transportation intervention. The theory, model, or framework did not need to explicitly use the term *implementation science* but rather include implementation related constructs.

After the database searches, Research Information Systems (RIS) files were imported into Covidence Systematic Review Software Version 2.0 (Veritas Health Innovation, Melbourne, Australia), an online review software that supports de-duplication and screening. NJ, RM, and MM independently screened each record's title and abstract against the inclusion and exclusion criteria. Any conflicts were resolved by discussion with SAM or DF. Full texts of included records were reviewed then screened by MM and NJ independently. Any conflicts were resolved by discussion with SAM or DF.

Table 1
Nilsen's (2015) five categories of theories, models, and frameworks used in implementation science.

Category	Description
Process models	Describing and/or guiding the process of translating research into practice
Determinant frameworks	Understanding and explaining what influences implementation outcomes
Classic theories	The different types of determinants specified in determinant frameworks can be linked to classic theories
Implementation theories	To achieve enhanced understanding and explanation of certain aspects of implementation
Evaluation frameworks	Provides a structure for evaluating implementation endeavors; action models

3.3. Data extraction

Once screening was complete, pre-determined data from each article were extracted in duplicate into Microsoft Excel tables by two reviewers, NJ and MM, independently. Extraction tables included study characteristics, such as each article's title, author, year, country, population, aim, context, and methods. Once extraction was complete, data from the two extraction files were compared and one final extraction table was finalized by consensus.

3.4. Data analysis

We used narrative synthesis to describe the patterns and themes resulting from the scoping review (Popay et al., 2006; Rodgers et al., 2009). Narrative synthesis organizes data, telling "a convincing story of why something needs to be done, or needs to be stopped, or why we have no idea whether a long established policy or practice makes a positive difference" (Popay et al., 2006, p. 5), aligning with our aim to highlight key barriers and facilitators to implementation across the included studies. Our synthesis included four elements: 1) theory development; 2) preliminary synthesis of included studies; 3) exploring the relationship within and between studies; and 4) assessing the robustness of the synthesis (Popay et al., 2006; Rodgers et al., 2009).

Second, we used thematic analysis to identify and organize themes (Braun & Clarke, 2014; Braun et al., 2017) relating to barriers and facilitators. Employing thematic analysis extends the narrative synthesis through an integrative approach to evidence synthesis (Dixon-Woods et al., 2005). We followed Braun and Clarke's (2012) six steps for thematic analysis: 1) familiarization of data; 2) generation of codes; 3) generating themes; 4) reviewing and confirming themes; 5) exploring theme significance; and 6) reporting findings.

Third, we used Nilsen's (2015) five categories of implementation theories to classify and discuss the theories, models, and frameworks we identified. Nilsen identifies five types of implementation theories: 1) process models; 2) determinant frameworks; 3) classic theories; 4) implementation theories; and 5) evaluation frameworks (Table 1). According to Nilsen, "it is important to recognize that these theories, models, and frameworks differ in terms of their assumptions, aims, and other characteristics, which have implications for their use" (p. 8). Thus, utilizing Nilsen's taxonomy is particularly relevant when examining a broad range of implementation theories, models, and frameworks relevant to active and sustainable transportation interventions.

4. Results

4.1. Results of the search strategy and selection processes

Fig. 1 presents the results of the article selection process. The searches resulted in 8892 records. After removing 331 duplicates (7 identified manually and 324 identified by Covidence), 8561 records remained for screening. Titles and abstracts of these 8561 records were

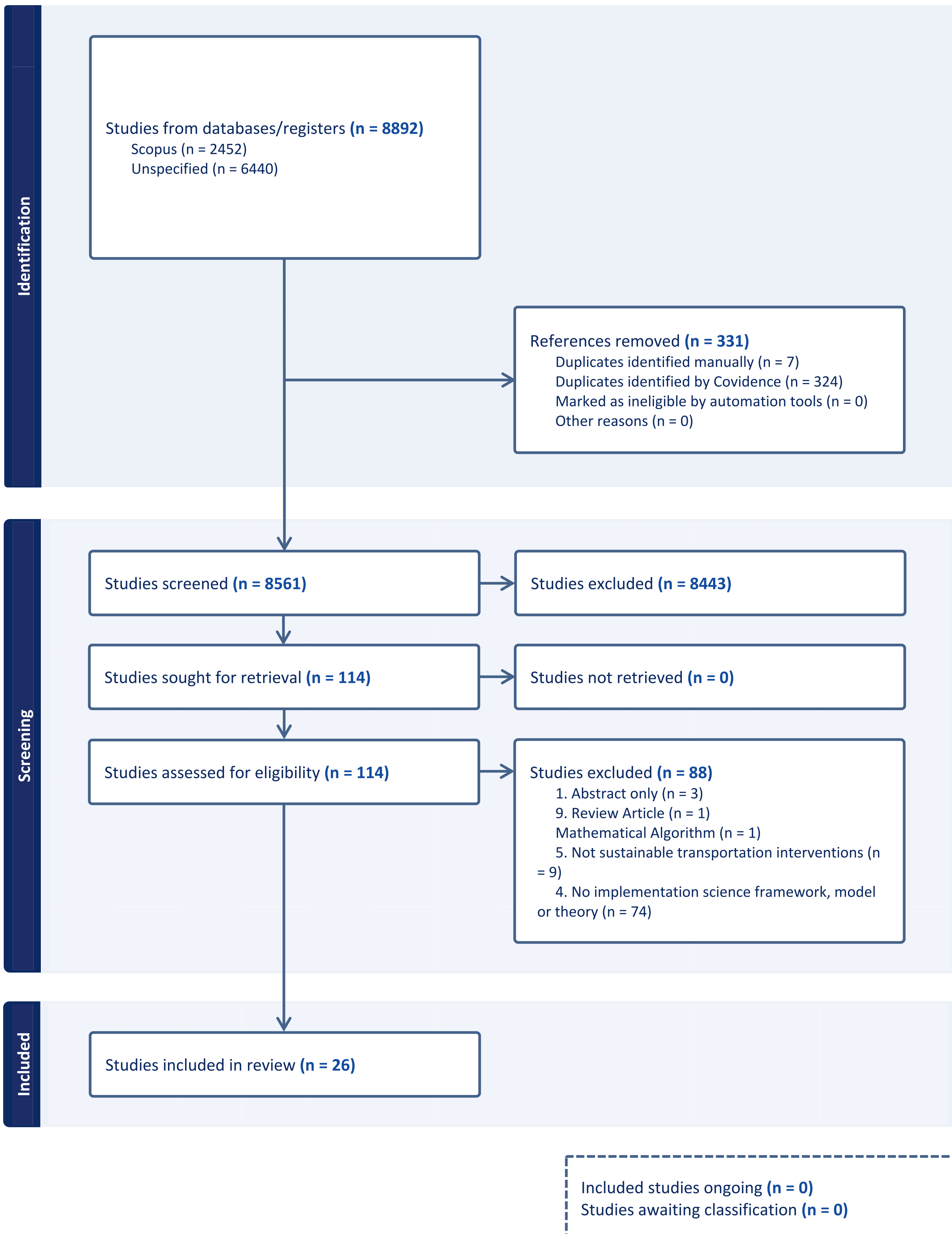

Fig. 1. PRISMA-ScR Flow Diagram – Results of article selection process.

Table 2
Publication details of extracted articles.

Title	Author(s)	Year	Country	Population	Aim	Context	Methods
Evaluation of the implementation of an intervention to improve the street environment and promote walking for transport in deprived neighbourhoods	Adams et al.	2017a	United Kingdom	Local residents	Evaluate implementation of Fitter For Walking (FFW) intervention and improve local neighbourhood walking environment	Deprived neighbourhoods	Interviews Focus groups Implementation log Participation records
Evaluation of the implementation of a whole-workplace walking programme using the RE-AIM Framework	Adams et al.	2017b	United Kingdom	Adults	Evaluate the implementation of Walking Works using the RE-AIM framework and provide recommendations for future delivery of whole-workplace walking programs	Five workplaces	Cross-sectional surveys Telephone interviews
Assessing the public participation initiatives in the planning and delivery of the Amman Bus Rapid Transit Project	Al-Sharari	2022	Jordan	Project sponsors, project team, contractors, and civic society groups	Assess the public participation initiative in the planning and delivery of Amman Bus Rapid Transit project	Public transit network	Policy review Case study approach (questionnaire, semi-structured interviews, focus groups)
Hot or not? The role of cycling in ASEAN megacities: Case studies of Bangkok and Manila	Bakker et al.	2018	Bangkok, Thailand and Manila, Philippines	Cyclists	Assess the current situation and progress in cycling, using Bangkok and Metropolitan Manila as case study cities, and to describe the necessary conditions for advancing the significance of cycling in tropical megacities	Middle-income countries	Systematic review of international and local literature
Implementation Strategies to Support Built Environment Approaches in Community Settings	Balis and Vincent	2023	Arkansas, USA	Extension Family and Consumer Science Agents	Inform implementation strategies through understanding delivery agents' perceptions of built environment approaches, a toolkit developed to support implementation, and other required implementation strategies	Communities in need of physical activity	Focus groups
Implementation of school remote drop-off walking programs: Results from qualitative interviews	Bejarano et al.	2021	California, Colorado, Florida, Minnesota, Michigan, North Carolina, and Wisconsin, USA	Key informant program leaders	Conduct a qualitative implementation evaluation of existing remote drop-offs across the U.S.	Elementary school surroundings	Interviews
Cycling and transition theories: A conceptual framework to assess the relationship between cycling innovations and sustainability goals	Bruno	2022	Netherlands and San Francisco, USA	Cyclists	Provide a framework for evaluating the relationship between cycling innovations and sustainability goals within a given institutional context using the Multilevel Perspective (MLP), Sustainable Mobility Paradigm (SMP), and Strategic Niche Management (SNM)	Institutional context	Integrated literature review Comparative case study approach
ELASTIC – A methodological framework for identifying and selecting sustainable transport indicators	Castillo & Pitfield	2010	England, UK	Transport planners at English municipal councils and transport-related academics at English universities	Address challenges associated with selecting appropriate sustainability indicators	Transportation planning sector	Surveys

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Table 2 (*continued*)

Title	Author(s)	Year	Country	Population	Aim	Context	Methods
Clear and consistent indicators to evaluate the effectiveness of sustainable urban transport plans	Chakhtoura & Pojani	2016	Paris, France	Parisian planners and politicians	Assess the extent to which transport sustainability targets have been achieved and whether existing evaluations have been adequate	Transportation planning sector	Transportation plan evaluation (internal evaluation studies, academic articles, press articles, online portals)
Challenges of promoting sustainable mobility on university campuses: The case of Eastern Mediterranean University	Dehghanmongabadi & Hoşkara	2018	Cyprus	EMU students and staff members	Promote active modes of transportation for commuting to and from, as well as transiting within, university campuses in response to the challenges of enhancing sustainability in the transportation sector	Eastern Mediterranean University (EMU)	Case study method (literature review, questionnaire survey, semi-structured interviews, photo-elicitation)
Planning accessible cities: Lessons from high quality barrier removal plans	Eisenberg et al.	2024	USA Cities	Municipal staff involved in developing and implementing Americans with Disabilities Act (ADA) transition plans	Identify facilitators for effective development and implementation of ADA transition plans for pedestrian infrastructure	West, mid-West, and Southern USA pedestrian environments	Semi-structured interviews
The International Universities Walking Project: Development of a Framework for Workplace Intervention Using the Delphi Technique	Gilson et al.	2009	Australia, Canada, England, the Netherlands, Northern Ireland, and Spain	Adults	Develop a consensual starting point for collaborative partners and a generic template for university walking interventions, which can be tailored to specific contexts	University workplace	Delphi technique (online questionnaire – 3 rounds)
Sustainable urban development in Riyadh: a projected model for walkability	Jarrar & Al-Homoud	2024	Curitiba, Brazil; Portland, USA; Copenhagen, Denmark; Tokyo, Japan	Pedestrians	Extract a Walkability Threaded Comprehensive Model from a critical comparison of case studies	Riyadh, Saudi Arabia	Case study comparative analysis
Framework for Assessing Public Transportation Sustainability in Planning and Policy-Making	Karjalainen & Juhola	2019	Helsinki, Finland and Toronto, Canada	Policy-makers	Present a Public Transportation Sustainability Indicator List (PTSIL) that provides a platform for an integrated assessment of environmental, economic, and social dimensions of sustainability through an indicator-based approach	Policy-making sector	Literature review and policy comparison (qualitative content analysis)
Development and implementation of Mobility-as-a-Service – A qualitative study of barriers and enabling factors	Karlsson et al.	2020	Finland and Sweden	Law-making and regional and local authorities	Empirically investigate how, and to what extent, different factors affect the development and implementation of MaaS	Public transportation	Multiple case studies (3 in Sweden, 1 in Finland)
Implementation contextual factors related to community-based active travel to school interventions: a mixed methods interview study	Koester et al.	2021	USA	School children, caregivers, and program coordinators	Identify contextual factors among low- and high-sustainability walking school bus programs in the USA	School transportation	Survey and semi-structured interviews
Towards people-centred integrated transport: A case study of Shanghai Hongqiao Comprehensive Transport Hub	Li & Loo	2016	Shanghai, China	Transit passengers	Establish a conceptual framework of integrated transport that includes different interchange performance indicators	Urban sustainable transport in developing countries	Observational survey and questionnaire
Pathways to integrating paratransit and	Mokoma & Venter	2023	Tshwane, South Africa	Public transport users	Ex-post assessment of the collaboration of actors when	Public transport integration projects	Literature review and case studies (documents, podcasts,

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Table 2 (*continued*)

Title	Author(s)	Year	Country	Population	Aim	Context	Methods
formal public transport: Case studies from Tshwane, South Africa					implementing integrated public transport		semi-structured qualitative questionnaire, interviews)
Does the service quality of urban public transport enhance sustainable mobility?	Mugion et al.	2018	Rome, Italy	Public transport users	Understanding the role of urban public transport service quality in increasing public transport use, reducing the adoption of private cars, and promoting intermodality.	Urban public transport systems	Territorial analysis, qualitative survey with in-depth interviews, and quantitative survey with structured questionnaire
Implementing bus rapid transit (BRT): A tale of two Indian cities	Rizvi & Sclar	2014	Ahmedabad and Delhi, India	Planners, residents, public transport users, and pedestrians	Demonstrate the role of the planning process in influencing BRT project outcomes	Public transport planning	Case comparison (published documents [i.e., meeting minutes, design and preparation documents, technical reports, newspaper articles, promotional pamphlets]; interviews; and meetings)
An Octopus and a circle at the basis of a framework for the evaluation of sustainable mobility	Sioui et al.	2018	High-level framework applicable to all settings	Transportation planners and policy-makers	Develop a sustainable mobility evaluation framework to guide planners in their decision process using a set of indicators with relevant estimation methods	Local, regional, and global sustainable mobility interventions	Literature review
Implementing a novel signage-only School Streets approach: Facilitators, barriers, and perceived outcomes	Todorova et al.	2025	Newcastle, England	School leaders, parents, pupils, and residents	Describe the facilitators and barriers to the implementation of signage-only School Streets as well as its perceived effectiveness	Four primary schools	Semi-structured interviews and focus groups
Incorporating Livability into Transportation Asset Management Practices through Bikeway Quality Networks	Vavrova & Chang	2019	San Francisco, USA	Cyclists	Describe a framework for implementing livability into transportation asset management (TAM) practices and to show how TAM systems can be enhanced by considering the needs of bicyclists through including bikeway maintenance and construction in a traditional pavement management system	Local transportation network decision-making	Condition assessment (bikeway pavement condition index [BPCI]), prioritization (asset inventory database and local municipality's bikeway plan), scenario analysis, and reporting
Socially sustainable transport in the context of different-sized cities in China: Conceptualisation and operationalisation of equity	Wan & Titheridge	2024	China	Disadvantaged mobility groups in China	Develop a notion of transport equity suited for the unique socio-economic, cultural, and political conditions in Chinese cities	Cities in China	Critical review of equity literature, analysis of Chinese transport planning and appraisal documents, semi-structured interviews with Chinese transport practitioners
Beyond the regime: can Integrated Sustainability Assessment address the barriers to effective sustainable passenger mobility policy?	Whitmarsh et al.	2009	Europe (Norfolk, United Kingdom)	Citizens and 'expert' European stakeholders (professionals with knowledge about transport systems and transport technologies [e.g., research/ academia, automotive sector, energy sector, NGOs, and government])	Considers whether Integrated Sustainability Assessment (ISA) offers an appropriate framework and toolkit to support better integration of sustainability into mainstream policy	Sustainability transport policy-making	Transition analysis (literature reviews) and stakeholder engagement (workshops, focus groups, questionnaires)

(*continued on next page*)

Table 2 (*continued*)

Title	Author(s)	Year	Country	Population	Aim	Context	Methods
Towards sustainable transport policy framework: A rail-based transit system in Klang Valley, Malaysia	Yusoff et al.	2021	Klang Valley, Malaysia	Policy-makers	Determine the key factors of a sustainable transportation framework and examine the key barriers in implementing a sustainable transport policy	Developing country	Case study on the development of railway lines (review of government reports and policies and in-depth interviews)

screened, resulting in 114 articles assessed at the full-text stage. Of these, 88 were excluded, most commonly because they did not employ an implementation science theory, model, or framework (n = 74). Other reasons for exclusion included abstract only (n = 3), not related to sustainable transportation interventions (n = 9), review article (n = 1), and mathematical algorithm (n = 1). In total, 26 studies were included in the final review.

Articles examined implementation frameworks applied to a range of active and sustainable transportation activities: walking (Adams et al., 2017a; 2017b; Bejarano et al., 2021; Eisenberg et al., 2024; Gilson et al., 2009; Jarrar & Al-Homoud, 2024; Koester et al., 2021; Todorova et al., 2025), cycling (Bakker et al., 2018; Bruno, 2022; Vavrova & Chang, 2019), walking and cycling (Balis & Vincent, 2023), and public transportation (Al-Sharari, 2022; Li & Loo, 2016; Mugion et al., 2018; Rizvi & Sclar, 2014; Yusoff et al., 2021), with several articles addressing multiple modes (Chakhtoura & Pojani, 2016; Castillo & Pitfield, 2010; Dehghanmongabadi & Hoşkara, 2018; Karlsson et al., 2020; Karjalainen & Juhola, 2019; Mokoma & Venter, 2023; Sioui et al., 2018; Wan & Titheridge, 2024; Whitmarsh et al., 2009). Table 2 presents the details of each article, including title, author, year, country, population, aim, context, and methods.

4.1.1. Nilsen's theories, models, and frameworks

Based on Nilsen's (2015) five categories of theories, models, and frameworks used in implementation science (Table 1), results show that many articles (n = 13) employed only one framework (Adams et al., 2017a; 2017b; Al-Sharari, 2022; Bakker et al., 2018; Dehghanmongabadi & Hoşkara, 2018; Eisenberg et al., 2024; Karjalainen & Juhola, 2019; Koester et al., 2021; Mokoma & Venter, 2023; Mugion et al., 2018; Sioui et al., 2018; Todorova et al., 2025; Vavrova & Chang, 2019), some (n = 11) used multiple frameworks, with several frameworks exhibiting characteristics of more than one category, and two studies (Jarrar & Al-Homoud, 2024; Wan & Titheridge, 2024) developed their own framework by examining existing literature relevant to their setting under study (Table 3).

An important finding is the broad range of implementation science frameworks used in active and sustainable transportation interventions, illustrating implementation complexity. For example, Vavrova and Chang (2019) utilized a process model, describing how research was translated into practice. Alternatively, Bakker and colleagues (2018) and Mugion and colleagues (2018) employed classical theory frameworks, using theories that originate outside of the implementation science field, to understand aspects of implementation. Interestingly, Jarrar and Al-Homoud (2024) developed a framework (Walkability Threaded Comprehensive Model), drawing from successful walkability case studies and synthesizing key features to fit their study setting of Riyadh, Saudi Arabia.

Several studies (n = 11) utilized more than one framework. For example, Balis and Vincent (2023) employed the Consolidated Framework for Implementation Research (CFIR) (Damschroder et al., 2009), a determinant framework, in concert with Proctor's Model (Proctor et al., 2011), an evaluation framework. Further, Whitmarsh and colleagues (2009) employed a sustainable transportation policy framework, a determinant framework and implementation theory. Gilson and colleagues (2009), Li and Loo (2016), and Yusoff and colleagues (2021) used a single framework, categorized as both a determinant framework and implementation theory. Overall, the range of frameworks illustrates that active and sustainable transportation interventions are complex, context-specific, and require multi-faceted approaches to implementation and evaluation.

Another important takeaway is that many studies (n = 12) employed a determinant framework, which "describe general types (also referred to as classes or domains) of determinants that are hypothesized or have been found to influence implementation outcomes" (Nilsen, 2015, p. 4). Given that implementation is greatly influenced by the context, it is unsurprising that determinant frameworks are used most frequently, as their flexibility allows for the determinants (classes or domains of barriers and facilitators) to be adapted to the particular setting. Further, Todorova and colleagues (2025) utilized the CIFR (Damschroder et al., 2022), a determinant framework, in an evaluative capacity post-implementation, which illustrates the versatility of determinant frameworks in the guiding and examination of implementation.

4.2. Narrative synthesis

We employed a narrative synthesis approach to organize findings from the included studies and identify barriers and facilitators to implementation, as well as examine the relationships between them. The first step, theory development, involves developing a theoretical model of how interventions work, why, and for whom (Popay et al., 2006). Throughout the articles we identified several salient factors influencing implementation that form the foundation for our theoretical model: 1) context; 2) community engagement and participation; 3) partnerships and collaboration; and 4) funding. These were further explored in our thematic analysis, allowing us to observe *which* factors could function as barriers or facilitators, depending on the setting and actors involved. In the following section we organize our findings to describe these barriers and facilitators (preliminary synthesis [step 2]), explore relationships within the data (step 3), and assess the robustness of the synthesis product (step 4).

4.2.1. Context

Accounting for context is a necessary condition for successful implementation (Edwards & Barker, 2014; Heath et al., 2012; Nilsen & Bernhardsson, 2019). The importance of assessing needs (Adams et al., 2017a; Balis & Vincent, 2023; Vavrova & Chang, 2019) and accounting for geographic differences (Al-Sharari, 2022; Bruno, 2022; Chakhtoura & Pojani, 2016; Jarrar & Al-Homoud, 2024; Mokoma & Venter, 2023;

Table 3
Extracted frameworks organized according to Nilsen's categories.

Title	Author(s)	Year	Framework	Framework Type
Evaluation of the implementation of an intervention to improve the street environment and promote walking for transport in deprived neighbourhoods	Adams et al.	2017a	Factors influencing implementation process (Durlak & DuPre, 2008)	Evaluation Framework
Evaluation of the implementation of a whole-workplace walking programme using the RE-AIM Framework	Adams et al.	2017b	RE-AIM	Evaluation Framework
Assessing the public participation initiatives in the planning and delivery of the Amman Bus Rapid Transit Project	Al-Sharari	2022	Appraisal Framework for Evaluating Public Participation Strategies in Developing Countries	Determinant Framework
Hot or not? The role of cycling in ASEAN megacities: Case studies of Bangkok and Manila	Bakker et al.	2018	Technological Innovation Systems (TIS) Framework (Bergek et al., 2008)	TIS: Classic Theory (based on Theoretical School of Innovation Systems)
			Multilevel Perspectives Framework for Socio-technical Transitions (Geels, 2012)	MLP: Classic Theory (based on evolutionary economics, the sociology of innovation, and institutional theory)
			Practice Theory (Watson, 2012)	Practice Theory: Classic Theory (based on theories of practice)
Implementation Strategies to Support Built Environment Approaches in Community Settings	Balis & Vincent	2023	CFIR (Damschroder et al., 2009)	CFIR: Determinant Framework
			Technology Acceptance Model (TAM) (Kwan et al., 2019; Lee et al., 2003)	TAM: Classic Theory (based on Theory of Reasoned Action)
Implementation of school remote drop-off walking programs: Results from qualitative interviews	Bejarano et al.	2021	CFIR (Damschroder et al., 2009)	CFIR: Determinant Framework
			Proctor's Model (Proctor et al., 2011)	Proctor's Model: Evaluation Framework
Cycling and transition theories: A conceptual framework to assess the relationship between cycling innovations and sustainability goals	Bruno	2022	Multilevel Perspective Framework (MLP) (Fraedrich et al., 2015; Köhler et al., 2019)	MLP: Classic Theory (based on evolutionary economics, the sociology of innovation, and institutional theory)
			Sustainable Mobility Paradigm (SMP) (Banister, 2008)	SMP: Determinant Framework & Evaluation Framework
			Strategic Niche Management (SNM) (Smith & Raven, 2012)	SNM: Classic Theory (based on sociology of innovation and evolutionary economics)
ELASTIC – A methodological framework for identifying and selecting sustainable transport indicators	Castillo & Pitfield	2010	ELASTIC (Evaluative and Logical Approach to Sustainable Transport Indicator Compilation)	Process Framework & Determinant Framework
Clear and consistent indicators to evaluate the effectiveness of sustainable urban transport plans	Chakhtoura & Pojani	2016	Sustainable Transportation Indicators	Implementation Theory & Evaluation Framework
Challenges of promoting sustainable mobility on university campuses: The case of Eastern Mediterranean University	Dehghanmongabadi & Hoşkara	2018	Transportation Demand Management (TDM) (Ferguson, 1990)	Evaluation Framework
Planning accessible cities: Lessons from high quality barrier removal plans	Eisenberg et al.	2024	CFIR (Damschroder et al., 2009)	Determinant Framework
The International Universities Walking Project: Development of a Framework for Workplace Intervention Using the Delphi Technique	Gilson et al.	2009	Consensus Framework (unique interlink framework)	Determinant Framework & Implementation Theory
Sustainable urban development in Riyadh: a projected model for walkability	Jarrar & Al-Homoud	2024	Walkability Threaded Comprehensive Model	Determinant Framework
Framework for Assessing Public Transportation Sustainability in Planning and Policy-Making	Karjalainen & Juhola	2019	Public Transportation Sustainability Indicator List (PTSIL)	Evaluation Framework
Development and implementation of Mobility-as-a-Service – A qualitative study of barriers and enabling factors	Karlsson et al.	2020	Analytical Framework for MaaS (Mukhtar-Landgren et al., 2016)	Classic Theory (based on institutional theory) & Implementation Theory
Implementation contextual factors related to community-based active travel to school interventions: a mixed methods interview study	Koester et al.	2021	CFIR (Damschroder et al., 2009)	Determinant Framework
Towards people-centred integrated transport: A case study of Shanghai Hongqiao Comprehensive Transport Hub	Li & Loo	2016	Integration Ladder	Determinant Framework & Implementation Theory
Pathways to integrating paratransit and formal public transport: Case studies from Tshwane, South Africa	Mokoma & Venter	2023	Simplified Framework for Assessing the Collaboration of Actors	Evaluation Framework
Does the service quality of urban public transport enhance sustainable mobility?	Mugion et al.	2018	Theoretical Model	Classic Theory (based on service quality in public transport theories)
Implementing bus rapid transit (BRT): A tale of two Indian Cities	Rizvi & Sclar	2014	Three Dimensional Framework	Process Model & Evaluation Framework

(continued on next page)

Table 3 (*continued*)

Title	Author(s)	Year	Framework	Framework Type
An Octopus and a circle at the basis of a framework for the evaluation of sustainable mobility	Sioui et al.	2018	Octopus Framework	Evaluation Framework
Implementing a novel signage-only School Streets approach: Facilitators, barriers, and perceived outcomes	Todorova et al.	2025	CFIR (Damschroder et al., 2022)	Determinant Framework
Incorporating Livability into Transportation Asset Management Practices through Bikeway Quality Networks	Vavrova & Chang	2019	The Bikeway Quality Framework	Process Model
Socially sustainable transport in the context of different-sized cities in China: Conceptualisation and operationalisation of equity	Wan & Titheridge	2024	Modified Conceptual Framework for Transport Equity in China	Classic Theory (based on three essential components of equity [Behbahani et al., 2019; Martens et al., 2019; McDermott et al., 2013]) & Determinant Framework
Beyond the regime: can Integrated Sustainability Assessment address the barriers to effective sustainable passenger mobility policy?	Whitmarsh et al.	2009	Integrated Sustainability Assessment (ISA) framework (Weaver & Rotmans, 2006)	Classic Theory (based on transitions and social learning theories) & Evaluation Framework
Towards sustainable transport policy framework: A rail-based transit system in Klang Valley, Malaysia	Yusoff et al.	2021	Sustainable Transportation Policy Framework for Railway	Determinant Framework & Implementation Theory

Wan & Titheridge, 2024; Whitmarsh et al., 2009) were identified in several articles. In their examination of implementation strategies to support built environment approaches for physical activity (e.g., walking and cycling), Balis and Vincent (2023) described the importance of learning about barriers and facilitators from ‘agents’ (i.e., potential beneficiaries of the implementation strategies) because it “is essential to ensure fit with needs, resources, and perceptions of what would work in the ‘real world’” (p. 503). Bruno (2022) used sustainability transitions theories (Banister, 2008; Köhler et al., 2019; Smith & Raven, 2012) to examine and compare cycling interventions (roundabouts) and showed how “one general type of cycling infrastructure can result in multiple possible outcomes for sustainability transition goals” (p. 2), depending on the geographic considerations.

Al-Sharari (2022) highlighted the importance of accounting for contextual differences, particularly in developing countries: “it is imperative for any reforms to consider the context of their own society. Often in developing countries, technocrats implement western ideologies, which may not result in changes that are suitable for developing nations” (p. 1316). This is echoed by Wan and Titheridge (2024) who claim that “a context-sensitive framework for conceptualizing equity is needed” (p. 2), due to differences in socio-economic characteristics, transport systems, built environments, and culture. In addition, Jarrar and Al-Homoud (2024) stressed the importance of recognizing “the unique physical, environmental, social, and economic characteristics of an urban setting” (p. 411) when implementing pedestrian infrastructure to enhance walkability.

The importance of contextual considerations was also demonstrated in articles examining sustainable transportation indicators (Castillo & Pitfield, 2010; Chakhtoura & Pojani, 2016; Karjalainen & Juhola, 2019). Castillo and Pitfield (2010) presented the ELASTIC framework for selecting indicators to evaluate sustainable transport systems and noted, “the objectives of sustainable transport are likely to differ with context, locality, time scientific knowledge, and global events” (p. 182), emphasizing that criteria are not pre-determined, only guided, by the ELASTIC framework. Chakhtoura and Pojani (2016) asserted that the choice of indicators depends on the scale of sustainable transportation plans, as well as the “geographical, economic, environmental, and social context for which they have been prepared” (p. 16). Further, Karjalainen and Juhola (2019) stated, “transportation sustainability assessments group indicators differently, depending on the approach, local contexts, and data restrictions, which in turn lead to differently operationalized sustainability dimensions” (p. 3).

4.2.2. *Community engagement and participation*

Community engagement and participation was a significant theme, particularly with regards to community buy-in and ongoing support (Bakker et al., 2018; Balis & Vincent, 2023; Bejarano et al., 2021; Koester et al., 2021; Mokoma & Venter, 2023; Todorova et al., 2025; Wan & Titheridge, 2024) because “involving local communities empowers them as active agents; fosters a sense of social responsibility, ownership, and belonging; and enhances social cohesion and sense of community” (Jarrar & Al-Homoud, 2024, p. 412). Koester and colleagues (2021) explained how engaging students, caregivers, and community members “were among the factors that emerged as most critical for supporting walking school bus program implementation” (p. 8). Bakker and colleagues (2018) asserted that a necessary component of cycling policy is citizen participation, which “includes not only active participation (e.g., user groups, activists), but also education and promotion of cycling activities” (p. 418), and was affirmed by Balis and Vincent (2023), emphasizing the importance of interest-holders’[1] participation for successful implementation. In addition, Bejarano and colleagues (2021) identified community support as a facilitator: “having community support (e.g., organizations, neighbourhood residents), resources, and partnerships were reported as helpful for program implementation and coordination” (p. 7). However, in the case of Todorova and colleagues' (2025) study “disjointed communication about launch dates and follow-up support led to a decrease in perceived buy-in from the parents and residents and concerns from school leaders” (p. 5).

Continuous community engagement was also identified as a facilitator (Adams et al., 2017a; Al-Sharari, 2022; Todorova et al., 2025). Adams and colleagues (2017a) described how consulting with the local community group to select a route or area of focus for their project was a facilitator and “checking-in” with the community program coordinators at regular intervals “is a strength of the research and helped to assess the development of the intervention as it evolved” (p. 18). Al-Sharari (2022) makes a similar claim, asserting that consultation with community partners and citizens during the planning and appraising stages of infrastructure projects is helpful for understanding their needs, information that might not be available to experts (Lee et al., 2013; Natarajan et al., 2019). Al-Sharari (2022) also noted that public participation takes

[1] The term “stakeholders” is used in the original article. We have changed the term to “interest-holders” to refrain from reproducing colonial language that may be perceived as disrespectful to Indigenous Peoples (Akl et al., 2024).

time and resources, and "the time needed to design participatory strategies, education, and mobilize participants may be too long for project owners as they may face time and budgetary constraints" (p. 1316).

Li and Loo (2016) undertook a person-centred approach to understanding passengers' public transportation travel experiences. Integrated transport systems are complex and involve diverse partners. Thus, it is necessary to understand passengers' perceptions of multimodal integration, particularly when attempting to overcome institutional barriers, such as extensive modal separation, funding shortfalls, insufficient staff for intermodal management, and failure to implement integrated fares and ticketing (Mulley & Moutou, 2015). Mugion and colleagues' (2018) study, exploring the role of service quality in peoples' sustainable transport decisions, also centres the experiences of users. Their results showed how citizens' perceptions of service quality and the overall satisfaction with public transport "can represent a driver for promoting the use of public means of transportation" (p. 1580). Lastly, some results affirmed the importance of gathering feedback through participatory planning processes (Rizvi & Sclar, 2014), emphasizing community feedback as a requirement for integrated sustainability assessment (Whitmarsh et al., 2009), and asserting that public engagement is required for developing successful sustainable transportation infrastructure (Yusoff et al., 2021). Overall, community participation can "foster trust, ownership, and learning" (Whitmarsh et al., 2009, p. 978) throughout, and after, the implementation process.

4.2.3. *Partnerships and collaboration*

Partnerships and collaboration were identified as an important facilitator (Todorova et al., 2025). For programs designed to facilitate physical activity, senior management support in work settings (Adams et al., 2017b; Gilson et al., 2009), and the support of school principals and/or school administration in school settings (Bejarano et al., 2021; Koester et al., 2021), was deemed crucial. Adams and colleagues (2017b) explained, "lack of senior management involvement and insufficient support on delivery of activities were mentioned as challenges for the walking champions in undertaking their role" (p. 11). Further, Gilson and colleagues' (2009) development of a framework to promote walking in university settings "represents a consensual starting point for collaborative partners" (p. 524), that can be used to guide physical activity interventions in other settings.

Dehghanmongabadi and Hoşkara (2018) recommended increasing collaboration between sustainable transportation councils and local organizations "to make decisions promoting successful sustainable transportation strategies" and "to encourage active transportation" (p. 15). Collaboration between university administration and government agencies is necessary to establish price control mechanisms for public transit. Further, the implementation and maintenance of sustainable transportation systems require partnerships and collaboration amongst transport companies (Mugion et al., 2018), similar to coordinating bikeway maintenance with existing infrastructure projects (Vavrova & Chang, 2019), which were described as facilitators. Lastly, Mokoma and Venter's (2023) examination of public transport integration projects showed that collaboration between formal public transport[2] and paratransit[3] organizations is necessary for successful implementation.

4.2.4. *Funding*

Not surprisingly, funding was identified as a facilitator and as a barrier (Adams et al., 2017a; 2017b; Balis & Vincent, 2023; Koester et al., 2021; Mokoma & Venter, 2023; Todorova et al., 2025; Yusoff et al., 2021). Results from Balis and Vincent's (2023) study showed that mini-grants (available up to $5,000 USD) helped to implement built environment strategies and "the goals of funding small-scale changes or pop-up events is to build community buy-in through small "wins" and provide evidence for funding larger projects" (p. 510). Bejarano and colleagues' (2021) study described the remote drop-off program as "being feasible to run and maintain" because it "did not take a substantial amount of work, time, or funds to initiate" (p. 6). Further, results of Gilson and colleagues' (2009) study showed that it was critical to build research capacity to examine physical activity program implementation through external funding, which "would promote free access and facilitate participant involvement" (p. 522). Results from Karlsson and colleagues' (2020) study also showed that the implementation of Mobility-as-a-Service, a form of integrated sustainable transport, requires funding, which "could come from public or private actors, or most likely a combination" (p. 292).

Regarding integrated urban sustainable transport systems, results from Mugion and colleagues' (2018) study showed "car-sharing is currently perceived as an expensive solution, even if people are interested to use it more" (p. 1579), showing how a lack of funding is a barrier. Yusoff and colleagues (2021) found "the viability of the projects is based on demands, financial capability, and stability" (p. 16), drawing attention to the importance of "financial readiness" (p. 23) as a facilitator. This is echoed by Vavrova and Chang (2019) in their study incorporating transportation asset management (TAM) into a bike quality framework: "because of the financial constraints that transportation agencies are facing, it is crucial to incorporate livability principles into the agency's existing processes" (p. 413). Lastly, Yusoff and colleagues (2021) identified cost as a main challenge for operating railway transport due to transport flows routed through congested railway infrastructure, emphasizing the need for more funding dedicated towards sustainable transportation implementation and evaluation.

4.3. *Equity*

In this final results section, we highlight findings relevant to transportation equity across the extracted studies (Table 4). To recall, transportation equity "encompasses the fair distribution of transportation resources, inclusive participation in decision-making processes, and recognition of the prevailing injustices that shape different levels of need and power within transportation systems" (Williams et al., 2023, p. 7). The majority of included studies (n = 17) did not address equity, a notable gap which we will discuss briefly before focusing on articles that *did* account for equity in various ways (n = 9).

4.3.1. *Equity excluded*

Adams and colleagues (2017b) utilized the RE-AIM framework to examine the implementation of a Walking Works program, which does not include equity-focused constructs. The CFIR (Damschroder et al., 2009) was utilized to evaluate a physical activity toolkit (Balis & Vincent, 2023), the implementation of a school drop-off program (Bejarano et al., 2021), and contextual factors related to community-based active travel to school interventions (Koester et al., 2021). However, the version of the CFIR used here does not have any equity- or accessibility-focused constructs.[4] Further, several studies employed the language of accessibility, as well as equity (Balis & Vincent, 2023; Gilson et al., 2009); however, while accessibility is integral to equity, it is often used to refer to physical access to infrastructure, such as wide sidewalks, shading devices, and lighting (Jarrar & Al-Homoud, 2024), for able-

[2] Formal public transport includes fixed routes, scheduled bus and train services operated by formally regulated and managed organizations (Mokoma & Venter, 2023).

[3] Paratransit refers to a flexible mode of public transport that does not follow a fixed schedule and typically uses small to medium-sized buses (Schalekamp et al., 2015).

[4] It is important to note that since the studies were conducted an updated version of the CFIR was developed (Damschroder et al., 2022) that includes constructs such as "human equality-centredness" within the Inner Setting domain, which more concretely centres equity within the framework.

Table 4
Equity constructs and indicators across frameworks.

Author(s) & Year	Framework	Construct/Indicator/ Dimension	Definition
Adams et al. (2017a)	Factors influencing implementation process (Durlak and DuPre, 2008)	Research scope: encouraging walking in socio-economically deprived neighbourhoods	
Castillo & Pitfield (2010)	ELASTIC	Equity and social inclusion	A sustainable transport system should contribute to both social and spatial equity by meeting the basic mobility and accessibility needs of all social, economic and geographical groups
Chakhtoura & Pojani (2016)	Sustainable Transportation Indicators	Public Transport Modes	Public transport frequency of service (by location) Public transport punctuality/ reliability (by location) Public transport vehicle/ station quality (heating/ air conditioning, cleanliness, overcrowding/ image, etc.)
		Non-motorize Modes	Length of cycling lanes (segregated/ non-segregated) Provision of bicycle parking facilities by land use (residential/ commercial) and type (open/ sheltered) Provision of bicycle service points (area coverage) Signal timing in favour of cyclists/ pedestrians/ public transport Traffic calming zones Quality of pedestrian paths, crossings, etc.
		Mobility of Vulnerable Groups	Provisions for the mobility of children/ older adults/ persons with disabilities Public transport vehicle/ station accessibility
		Accessibility and Wayfinding	Clarity/frequency of signage and information systems (electronic/conventional)
		Affordability	Public transit affordability (expenditures as percentage of household budget/ subsidies for vulnerable groups, etc.)
Eisenberg et al. (2024)	CFIR (Damschroder et al., 2009)	Research scope: removing access barriers in pedestrian infrastructure for persons with disabilities	
Karjalainen & Juhola (2019)	PTSIL	Social Sustainability – Physical Access	Increase proximity to service Increase service frequency Decrease travel time Increase multi-modality Increase inter-modality Increase access for persons with disabilities
		Social Sustainability – Socio-economic Status	Decrease share of income used on public transit Increase attention to older adults, young, and low-income
		Social Sustainability – Information Availability	Increase timely service information availability Increase personal journey planner availability Increase transparency in decision-making
		Social Sustainability – Attractiveness	Increase user satisfaction Increase active promotion to non-users Increase service reliability
		Social Sustainability – Planning and Participation	Increase public participation
Li & Loo (2016)	Integration Ladder	Walking Distance	Moving walkways Lifts
		Walking Environment	Air-conditioning Rain-shelter
		Waiting environment	Air-conditioning Rain-shelter Toilet for all kinds of users
		Signage	Multiple languages, acoustic signals, maps, directional signs, Braille maps
Sioui et al. (2018)	Octopus Framework	Society – Social Justice	Social equity Equal basis Help to disadvantaged groups
		Society – Quality of Life	Accessibility (quality of service)
Todorova et al. (2025)	CFIR (Damschroder et al., 2022)	Human equality-centredness	There are shared values, beliefs, and norms, about the inherent equal worth and value of all human beings
Wan & Titheridge (2024)	Modified Conceptual Framework for Transport Equity in China	Impacts to be distributed	Accessibility to jobs, education, and key services and culture-related activities Environment, safety, and health issues related to transportation Impact on forming community and social network Subjective wellbeing and quality of life Economic impacts, such as land acquisition and resettlement, and regional economic development Public participation in decision-making
		Target groups	Key groups identified by socio-demographic characteristics People without urban or local citizenship (Hukou) People living in specific housing/communities Disadvantaged families/households
		Distributive principles	Maximizing the development prospects of disadvantaged groups is the most appropriate principle as a general and long-term guidance for transport equity in China For some small- to medium-sized cities, a more practical principle is to apply a threshold for advantaged groups to ensure their basic needs are met and they are not further disadvantaged by a proposed intervention

bodied road users and does not tackle the complex dimensions of equity and inequity within implementation.

4.3.2. Equity included

Equity was accounted for in nine studies, either in the overall scope (Adams et al., 2017a; Eisenberg et al., 2024) or in the dimensions and/or constructs of the implementation theory, model, or framework (Table 4). For example, Wan and Titheridge (2024) examined the meaning of transportation equity in the context of China, inspired by the fact that the majority of current research describing and/or measuring transportation equity has been established and developed in Western countries (Zhang & Zhao, 2021) and is, thus, not helpful in China's transportation context. The authors assert "a context-sensitive framework for conceptualizing equity is needed" (Wan & Titheridge, 2024, p. 2) and, to achieve this, the authors conducted a literature review, policy document analysis, and semi-structured interviews with transport professionals working in China. They identified equity considerations, such as "accessibility to jobs, education, and key services and culture-related activities" (p. 9), as well as some general suggestions for achieving this: "maximizing the development prospects of disadvantaged groups is the most appropriate principle as a general and long-term guidance for transport equity in China" (p. 9).

Castillo and Pitfield's (2010) study examining the use of the ELASTIC framework to determine sustainable transport indicators includes "equity and social inclusion" as a general construct to evaluate context-specific indicators and their ability to reflect how a transportation system upholds such values; however, the results of this examination only assert the importance of equity considerations and do not offer any strategies for incorporating equity into transportation planning. Further, Adams and colleagues' (2017a) study examined implementation of the Fitter for Walking program in deprived areas, which prioritizes equity in its scope, but did not include any equity-focused constructs (Durlak & DuPre, 2008).

Chakhtoura and Pojani (2016) identified several sustainable transportation indicators that include equity considerations: public transport modes, non-motorized modes, mobility of vulnerable groups, accessibility and wayfinding, and affordability. These indicators are prescriptive, offering ways to ensure equity is accounted for in policy and decision-making. For example, the sustainability of *public transport modes* is measured by its frequency of service, punctuality and reliability, and vehicle/station quality (e.g., heating/air conditioning); *motorized modes* are measured by length of cycle lanes, provision of bicycle parking facilities, provision of bicycle service points, signal timing in favour of cyclists/pedestrians/public transport, traffic calming zones, and quality of pedestrian paths/crossings; and *mobility of vulnerable groups* is measured by provisions for the mobility of children/older adults/persons who are living with a disability, and public transport vehicle/station accessibility. However, while these indicators show *what* needs to be done to ensure equity is accounted for, they do not provide strategies for *how* to implement these interventions.

Similarly, Karjalainen and Juhola (2019) created an analytical framework of indicators for measuring the sustainability of public transportation services (PTSIL) by organizing findings from the literature across three sustainability domains: environmental, social, and economic. The social sustainability domain includes accessibility, defined as "equitable transportation that provides access to opportunities, reduces exclusion, and aims to increase the quality of life" (p. 5) and is associated with several equity-focused indicators: increased service frequency, decreased travel time, increased access for persons with disabilities, increased attention to older adults, young, and low-income, and increased public participation. The authors then performed a content analysis to show how these indicators are embedded in public transportation policy documents in Helsinki, Finland and Toronto, Canada. Several indicators align with those identified by Chakhtoura and Pojani (2016); however, Karjalainen and Juhola (2019) only examined which indicators are included in the policies reviewed and do not prescribe any strategies for translating policy to practice.

Li and Loo (2016) examined the effectiveness of multi-modal integration at a specific transit hub in Shanghai, China using a people-centred approach, which included an observational survey and questionnaire distributed to passengers. Based on these experiences, the authors developed a conceptual framework of interchange performance indicators referred to as the "integration ladder," which identified barriers to multi-modal integration across different levels of integration (high, moderate, low). Broadly, equity-focused indicators included moving walkways, lifts, air-conditioning, rain-shelter, toilets for all kinds of users, and signage (multiple languages, acoustic signals, maps, directional signs, Braille maps). To remove barriers preventing the implementation of these features, the authors assert the importance of institutional collaboration between governing bodies overseeing the different modes of transport, which often have different standards, practices, and funding sources.

Sioui and colleagues (2018) proposed a framework (the Octopus Framework) for sustainable transportation planners and decision-makers that included indicators across three dimensions: environment, society, economy. Within the social dimension of sustainability indicators included social justice for disadvantaged groups, and quality of life, with regards to accessibility. Further, the authors illustrated how indicators across the three categories influence and affect each other and propose an additional layer to this framework titled "the Causal Circle." For example, the *quality of life* indicator within the social dimension is affected by the *greenhouse gas emissions* indicator within the environment dimension. The authors claim that both policy-makers and transportation planners can adapt this framework to guide sustainable mobility efforts in their particular contexts.

Eisenberg and colleagues' (2024) study examined the effectiveness of removing existing infrastructure barriers that did not align with the Americans with Disabilities Act (ADA). The authors reflect on the plethora of National Disability Policies adopted, which are intended to reduce barriers for persons with disabilities; however, "while adopting policies is a critical and significant milestone, questions remain about the implementation of those policies" (p. 1). In other words, the authors are skeptical of whether policy adoption truly removes barriers for persons with disabilities, explaining that effective barrier-removal plans must include a clear schedule, identification of responsibilities, and communication with the public. Further, the plan must include specific dates and locations, funding sources, and be informed by priorities identified by persons with disabilities. To investigate further, the authors utilized an implementation science framework (the CFIR; Damschroder et al., 2009) to examine factors that led to successfully implementing ADA policies and interviewed municipal staff (ADA coordinators, project managers, engineers, etc.) across the west, mid-west, and southern United States who were involved in policies that were effective in removing such infrastructure. While the authors utilized the original version of the CIFR, which does not explicitly account for equity, results mapping onto constructs within the Intervention Characteristics and Outer Setting domains illustrated the importance of adapting the approach to the particular setting and understanding the specific needs of persons with disabilities living and traveling in those areas.

Lastly, Todorova and colleagues (2025) utilized the updated CFIR (Damschroder et al., 2022) to examine barriers and facilitators to implementing a School Streets[5] scheme. The updated CFIR includes explicit equity constructs, such as human-equality-centredness: "there are shared values, beliefs, and norms, about the inherent equal worth and value of all human beings" (p. 6). However, data from their interviews did not specifically map onto this construct. Interestingly, implicit equity considerations were identified in data mapping onto other

[5] School Streets is a targeted approach to increasing active travel to and from school via a time-specific restriction order for motor vehicles.

domains and constructs, such as the Engaging construct within the Implementation Process domain: "engagement activities must reflect the challenges of their specific area" (Todorova et al., 2025, p. 5), such as neighbourhoods impacted by social exclusion, perceived criminality, and violence.

The final stage of narrative synthesis involves assessing the robustness of the synthesis by drawing conclusions about general patterns of barriers and facilitators (Popay et al., 2006; Rodgers et al., 2009). Those identified illustrate complex factors that do not operate in isolation. Alternatively, context, community engagement and participation, partnerships and collaboration, and funding are inextricably linked and significantly influence successful implementation. Accounting for equity in this process is crucial. Implementation frameworks need to be adapted to the context in which they are applied, thus requiring equitable strategies to engage communities, collaborate and establish partnerships, and secure funding. In other words, incorporating community engagement and participation strategies into the implementation process for active and sustainable transportation interventions requires robust and meaningful uptake of equity considerations (Grossardt & Bailey, 2018; Ricci, 2021).

5. Discussion

This scoping review synthesized current literature on implementation science frameworks and their application in active and sustainable transportation interventions, focusing on transportation equity. A key takeaway is that such frameworks exist, as evidenced by the broad range of frameworks identified in this review that employed and/or were developed for particular active and sustainable transportation interventions, covering a wide range of contexts. The following discussion more fulsomely examines how equity was conceptualized and included in the papers extracted and situates our findings within the context of existing implementation science scoping reviews.

5.1. Transportation equity

This review sought to understand how transportation equity is addressed across implementation science frameworks for active and sustainable transportation (Maher et al., 2020; Martens et al., 2019; Shiller & Kenworthy, 2017). In the studies included, equity considerations highlight key relationships across the factors addressed in the narrative synthesis, which illustrate the complexity of the implementation landscape for active and sustainable transportation interventions. While seven frameworks explicitly addressed equity in their constructs or domains, two studies accounted for equity in the overall scope but did not use equity-focused constructs (Adams et al., 2017a; Eisenberg et al., 2024). For example, Eisenberg and colleagues' (2024) study illustrated how a framework (CFIR; Damschroder et al., 2009), when used to examine an equity-focused issue (the translation of national disability polices to practice) can elicit meaningful results about *how* equity was included, or not. This is an important result from this work; specifically, we need not rely on active and sustainable transportation frameworks to include equity, so long as the research question and process embeds equity from the start and throughout. However, if equity is lacking in the design and implementation of a particular intervention, an implementation framework that prioritizes equity throughout the process is crucial. Alternatively, Todorova and colleagues (2025) utilized the updated CIFR (Damschroder et al., 2022), which *does* include equity-focused constructs (e.g., human equality-centredness) onto which their data did not map. That being said, while some studies included constructs such as "equity and inclusion" (Castillo & Pitfield, 2010) and "mobility of vulnerable groups" (Chakhtoura & Pojani, 2016), most did not account for different user types or, if they did, the specific needs of each group were not identified, which is required given the range and sometimes conflicting needs of road users (McCullogh et al., 2023).

Equity concerns underpin the linkages between context-specific implementation strategies, community participation and engagement, and funding. Providing fair representation and participation opportunities to the public is an important feature of equitable implementation because without equal representation, effective participation in the sustainable transportation intervention is difficult to achieve (Marzuki, 2015; Karjalainen & Juhola, 2019; Wan & Titheridge, 2024) and will not reflect the needs of the community (Eisenberg et al., 2024; McCullogh et al., 2023; Whitmarsh et al., 2009). This concern is addressed by Al-Sharari (2022) who explained how traditionally excluded groups (e.g., those living under poverty, ethnic minorities, women) generally struggle to meet basic needs and can be "deterred from participating by a feeling that they lack education or knowledge, particularly when it comes to complex infrastructure projects" (p. 1316). Community engagement and participation were identified as facilitators to implementation; however, "groups might be actively ignored and marginalized by project proponents" and "excluding some from participating bequeaths unequal powers among stakeholders" (Al-Sharari, 2022, p. 1316), which is a barrier. This illustrates a failure to provide a comprehensive understanding of public views and results in an incomplete picture of the implementation context (Karjalainen & Juhola, 2019; Whitmarsh et al., 2009).

Car-oriented planning and prioritization in transportation has been established as a substantial barrier as well (McCullogh et al., 2022; Petzer et al., 2021; Woodcock & Aldred, 2008). Transportation infrastructure that prioritizes cars is an equity issue because many road users cannot afford a motor vehicle (Mattioli et al., 2017; Sallis et al., 2016). In this review, car-oriented planning was identified as a barrier to choosing sustainable modes of transport (Al-Sharari, 2022; Bakker et al., 2018; Jarrar & Al-Homoud, 2024; Mugion et al., 2018). For example, the prioritization of motor vehicles in urban planning and lack of active and sustainable transportation options maintained students' dependency on cars (Dehghanmongabadi & Hoşkara, 2018). Further, traffic-related concerns prevented parent buy-in to a school remote drop-off program (Bejarano et al., 2021). Car-oriented planning also has linkages with partnerships and collaboration because establishing integrated networks of car-sharing to promote single-use car trips requires collaboration across municipalities (Mugion et al., 2018).

Lastly, the work of Wan and Titheridge (2024) illuminated important connections between context and transportation equity; in particular, their review of equity literature showed that Western definitions and metrics of transportation equity (e.g., mobility/accessibility; traffic-related pollution; traffic safety; and health [Martens et al., 2019]) do not fully align with what is needed in the Chinese context: "transportation equity in China may need to focus more on resources and opportunities for people to contribute to community and national development" (Wan & Titheridge, 2024, p. 4). Such a distinction is based on *harmony*, an important concept in Chinese culture, which prioritizes balance between people and the environment, as well as the connections between people in a society (Yang, 2012). Further, the authors note that the Western categories of demographic characteristics (e.g., age, ethnicity, gender, education, income, culture, disability, etc.) do not fully capture the range of vulnerable groups living in China due to China's institutional structure, which is characterized by a population management strategy that emphasizes the rural–urban dichotomy and exacerbates transportation disadvantage for rural dwellers. Thus, in the Chinese context, the authors argue that transportation equity ought to include distinct supports for folks living in these areas.

5.2. Accounting for context in implementation

As noted above, accounting for contextual particularities when utilizing implementation theories, models, and frameworks to support the implementation of interventions (Karjalainen & Juhola, 2019; Wang et al., 2023) is pivotal. While our search parametres did not include specific terms such as "context-sensitive" or "context-specific," we acknowledge that these could yield additional results for future work.

Accounting for context "is necessary to explain how or why certain implementation outcomes are achieved, and failure to do so may limit the generalizability of study findings to different settings or circumstances" (Nilsen & Bernhardsson, 2019, p. 2). This is certainly the case with frameworks for implementing active and sustainable transportation interventions, as the results of this scoping review illustrate.

The importance of accounting for context is affirmed by the included studies and by Nilsen and Bernhardsson's (2019) scoping review of determinant frameworks describing contextual factors influencing implementation outcomes, which identified twelve context dimensions utilized throughout selected studies. Further, Wang and colleagues' (2023) scoping review of implementation science frameworks, models, and theories sought to establish a foundation for the systematic appraisal of their quality by assessing the usability, applicability, and testability of existing frameworks, models, and theories. Their results showed the majority of studies focused on the micro-level constructs relevant to the particular context, aligning with literature emphasizing the importance of context in implementation science (Kourtit & Nijkamp, 2022; Nilsen & Bernhardsson, 2019; Proctor et al., 2023; Wang et al., 2023). Our results align with these findings, illustrating the importance of context-specific implementation strategies; however, our results are novel as they prioritize equity as a necessary component of implementation theories, models, and frameworks and highlight notable gaps and future directions in the realm of active and sustainable transportation.

5.3. Policy guidance

Several policy implications were gleaned from our review, which can aid in the replication and scaling-up and scaling-out of interventions (Winters et al., 2024). Balis and Vincent (2023) recommend policymakers include implementation strategies in concert with policy mandates "to improve the translation of research to practice" (p. 510). Indicators were identified as "fundamental tools that direct policymakers in the creation and assessment of policy targets" (Chakhtoura & Pojani, 2016, p. 16), and in the context of sustainable transportation policy, indicators enable measurement of developments over time and space (Litman, 2023). Further, Karjalainen and Juhola (2019) recommends the application of a "standardized and more comprehensive set of indicators and policy objectives" (p. 15), such as the versatile PTSIL, similar to Sioui and colleagues' (2018) flexible 'Octopus Framework' that allows policymakers to adapt sustainable mobility strategies to their social context (time, place) and interests.

Community engagement in transportation policy design was identified as crucial (Adams et al., 2017a; Al-Sharari, 2022; Castillo & Pitfield, 2010; Jarrar & Al-Homoud, 2024; Whitmarsh et al., 2009; Yusoff et al., 2021); however, engagement must be incorporated throughout the implementation process and not solely during planning and appraisal stages. Further, the scope of public engagement should extend beyond those immediately affected by the intervention because such a limit "can lead to the implementation of rudimentary and front-loaded participatory practices" (Al-Sharari, 2022, p. 1318) or, in the case of the bus rapid transit proposal, the lack of active engagement with car users or attention paid to "the impact proposals would have on this powerful group" (Rizvi & Sclar, 2014, p. 203), thus, slowing down implementation. This aligns with Li and Loo's (2016) recommendation that "more people-centred terminal design and policymaking are needed for realizing seamless integration in the transport hub design" (p. 55), and Mugion and colleagues' (2018) findings indicating that public transportation service quality impacts decisions of road users to utilize public transportation infrastructure. Thus, policies aimed at enhancing service quality "can contribute to reducing the usage of private cars" (p. 1581).

More specific policy recommendations include modifying workplace policies regarding car parking arrangements, work hours, and dress codes to encourage walking to and from work (Adams et al., 2017b; Gilson et al., 2009), similar to modifying school drop-off locations to areas remotely around schools to encourage walking to school (Bejarano et al., 2021; Todorova et al., 2025). Further, law-making authorities can facilitate sustainable transportation, broadly, "by adjusting relevant regulations and policies such as transport-related subsidies, taxation policies, and the definition of public transport" (Karlsson et al., 2020, p. 293). Lastly, incorporating cycling into sustainable transportation systems was identified as a necessary step by policymakers, particularly in developing countries (Bakker et al., 2018; Bruno, 2022; Vavrova & Chang, 2019). In other words, policymakers need to create "adequate conditions for safe cycling on as many roads as possible" (Bakker et al., 2018, p. 430).

While a plethora of policy implications were described, prioritizing equity considerations in active and sustainable transportation were only addressed in nine articles (Table 4), where they were included at the policy and planning levels without any strategies to ensure their application to local settings. As Eisenberg and colleagues' (2024) study demonstrated, an implementation science framework (e.g., the CFIR; Damschroder et al., 2009) that does not include equity but is used to examine and equity-focused intervention, can meaningfully illustrate equity inclusion or exclusion in implementation. However, if the intervention is not equity-focused, it is integral that the implementation science framework employed includes explicit and meaningful equity-focused constructs. Thus, implementation theories, models, and frameworks designed to facilitate active and sustainable transportation interventions ought to include input from the variety of community partners and road users to ensure an equitable transportation system (Williams et al., 2023). Incorporating engagement mechanisms and indicators into transportation proposals and planning is one avenue for policymakers to advance transportation equity (Cascetta & Pagliara, 2013; Syal et al., 2024).

6. Strengths and Limitations

To our knowledge, this scoping review is the first to summarize implementation science theories, models, and frameworks applied to active and sustainable transportation interventions. We explored key terms related to implementation science; however, we recognize this area of study includes diverse disciplines that use different implementation science terminologies. Further, we acknowledge that there is a breadth of literature examining the implementation of active and sustainable transportation interventions that does not use the language of implementation science or any formal theory, model, framework (e.g., Malik et al., 2020; Wegman et al., 2023). This body of research can offer meaningful contributions to the examination of implementation science in the context of active and sustainable transportation but falls outside of the scope of this current review.

While we carefully developed our search strategy and had it reviewed by content experts, we recognize that our key term list may not have included all relevant terms. Further, while the search identified several studies including sustainable transportation interventions, the search did not explicitly list intervention types, and there may be relevant literature that was excluded. Finally, our review did not appraise the quality of the articles included; although, given the type of data we extracted, appraisal may be less relevant and is not standard in scoping review methodology.

7. Future directions

Future directions for research examining implementation science for active and sustainable transportation include, 1) an identification and examination of the different types of interventions included in active and sustainable transportation that use implementation science frameworks; 2) development of indicators and mechanisms specifically for equitable community and partner engagement throughout the whole process of implementation; 3) examining the roles and responsibilities of people involved in the implementation process (e.g., politicians, engineers, community groups) and how active and sustainable transportation policies inform their work; and 4) develop a framework specifically for active and sustainable transportation interventions that prioritizes equity and can be adapted to particular contexts.

Developing a framework specifically for implementing active and sustainable transportation interventions is of particular importance due to the range of contextual particularities that influence implementation, as demonstrated by the results of our scoping review. These findings align with the CapaCITY/É research programme objectives identified above (Winters et al., 2024), the primary motivation for conducting this review.

Given the priority of context-specific approaches to implementation and how this influences, and is influenced by, community engagement, partnerships and collaboration, and funding, future directions for developing implementation requires complex multi-faceted strategies. Frameworks extracted were relevant to all five of Nilsen's (2015) categories (Table 3), demonstrating the context-specific nature of the implementation landscape for active and sustainable transportation interventions. Articles focusing on identifying indicators were not only explicit in their acknowledgement of accounting for context, but were also the frameworks in which equity was most explicitly addressed (Table 4). Future development of implementation science frameworks ought to include an indicator identification and measurement component. As stated by Castillo and Pitfield (2010), indicators have the "ability to capture the multidimensionality of sustainable transport and break down the complex concept into small and manageable units of information" (p. 181).

As noted above, implementing active and sustainable transportation interventions across diverse contexts is complex. One approach to developing an implementation science framework that captures these complexities is to draw lessons from research on sustainability transitions theories (Geels & Schot, 2007; Köhler et al., 2019), which focus on the interaction between the innovation (intervention), the socio-technical landscape in which the innovation is being implemented (the regime), and the actors involved (Köhler et al., 2019). Options include the Technological Innovation Systems framework (Bergek et al., 2008) and Multi-level Perspective (Geels, 2012) utilized by Bakker and colleagues (2018) and the Sustainable Mobility Paradigm (Banister, 2008) and Strategic Niche Management (Raven et al., 2010) utilized by Bruno (2022). Given that "mobilities are both productive of social relations [that involve the production and distribution of power] and produced by them" (Cresswell, 2010, p. 21), examining the implementation of active and sustainable transportation interventions through the lens of sustainability transitions is a promising future direction.

8. Conclusion

This review examined implementation science theories, models, and frameworks for active and sustainable transportation, with a focus on equity. Results show that there *is* published literature describing implementation science theories, models, and frameworks that have been utilized for implementing active and sustainable transportation interventions; however, there is not yet a generalizable, overarching, and flexible framework for these interventions. Given the importance of context-specific approaches identified in this study, more research is required to determine if an overarching framework for implementing active and sustainable transportation interventions is possible. The barriers and facilitators identified in the articles extracted provide *some* guidance for implementation best practices; however, to truly improve the health and mobility of diverse urban populations, equity must be central and embedded from the start of the implementation process.

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CRediT authorship contribution statement

Emily McCullogh: Writing – review & editing, Writing – original draft, Formal analysis. **Sarah A. Moore:** . **Rebecca Maher:** Investigation, Data curation. **Nazanin Jannati:** Writing – review & editing, Investigation, Data curation. **Maud Muosieyiri:** Investigation, Data curation. **Alison Macpherson:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Ben Beck:** Writing – original draft, Methodology, Funding acquisition, Conceptualization. **Daniel Fuller:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Search strategy example (Scopus)

#	Query
1	sustainab* W/2 (transport* OR mobility)
2	(green OR greening) W/1 (transport* OR mobility OR corridor* OR path*)
3	{bus rapid transit} OR brt OR {light rail transit} OR lrt
4	{public transit} OR "public transport*" OR "active transport*" OR "active travel*"
5	traffic W/1 (calm* OR restrain*)
6	(speed* W/1 (reduc* OR lower*)) AND (bike* OR bicycl* OR cycling OR cyclist* OR pedestrian*)
7	(bike* OR bicycl* OR cycling OR cyclist* OR pedestrian* OR walking) W/3 (lane* OR path* OR trail* OR network* OR infrastructure OR friendly)
8	walkab* OR bikeab*
9	(adaptive OR assistive) W/1 (bike* OR bicycl* OR cycling OR mobility OR transport*)
10	#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9
11	(implement* OR integration) W/15 (framework* OR model* OR concept* OR theor* OR fidelity)
12	{implementation science} OR {implementation research}

(continued on next page)

(*continued*)

#	Query
13	operationali* OR "scale up" OR "scale out" OR upscaling
14	knowledge W/1 (application OR broke* OR creation OR diffus* OR disseminat* OR exchang* OR implement* OR management OR mobili* OR translat* OR transfer* OR uptak* OR utili*)
15	evidence* W/1 (exchang* OR translat* OR transfer* OR diffus* OR disseminat* OR exchang* OR implement* OR management OR mobil* OR uptak* OR utili*)
16	KT W/1 (application OR broke* OR diffus* OR disseminat* OR decision* OR exchang* OR implement* OR intervent* OR mobili* OR plan* OR policy OR policies OR strateg* OR translat* OR transfer* OR uptak* OR utili*)
17	research* W/1 (diffus* OR disseminat* OR exchang* OR transfer* OR translation* OR application OR implement* OR mobil* OR transfer* OR uptak* OR utili*)
18	{research findings into action} OR {research to action} OR {research into action} OR {evidence to action} OR {evidence to practice} OR {evidence into practice}
19	diffusion W/1 innovation*
20	("systematic review*" OR "knowledge synthes*" OR "evidence synthes*") W/4 ("decision mak*" OR "policy mak*" OR "policy decision*" OR "health polic*")
21	("systematic review*" OR "knowledge synthes*" OR "evidence synthes*") W/1 (application OR implement* OR utili*)
22	"research utili*ation"
23	("evidence base*" OR "evidence inform*") W/1 (decision* OR plan* OR policy OR policies OR practice OR action*)
24	#11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23
25	framework* OR model* OR theor* OR concept*
26	#24 AND #25
27	#11 OR #26
28	#10 AND #27
29	#28 LIMIT-TO (LANGUAGE,"English")
30	#29 (EXCLUDE (DOCTYPE, "cp") OR EXCLUDE (DOCTYPE, "ch") OR EXCLUDE (DOCTYPE, "cr") OR EXCLUDE (DOCTYPE, "bk"))

Data availability

No data was used for the research described in the article.

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